Assignment 2

Data Acquisition and Survey Methods
Survey Analysis Report

Group 39

2026

Contents

Introduction

This report analyzes the relationship between meal habits and perceived study productivity among university students. The focus is on three research questions related to breakfast frequency, the number of main meals per day, and self-reported study productivity.

The report addresses the following three research questions and hypotheses:

1. **RQ1:** Is the frequency of eating breakfast associated with students' perceived study productivity?
   H1: Students who eat breakfast more often report higher perceived study productivity.

2. **RQ2:** Is the number of main meals per day associated with students' perceived study productivity?
   H2: Students who eat more main meals per day report higher perceived study productivity.

3. **RQ3:** Is the frequency of eating breakfast associated with the number of main meals students usually eat per day?
   H3: Students who eat breakfast more often also report eating more main meals per day.

For the analysis, only the variables relevant to these research questions were selected from the full survey dataset. These include three demographic variables and the three Group 39 survey variables on breakfast frequency, main meals per day, and perceived study productivity.

Data Preparation

```r
# Load the survey data and keep the original column names
data <- read.csv("../data/Survey_results_2026.csv", check.names = FALSE)

# Select only the columns relevant for our group
group39_data <- data[, c(
  "DQ How old are you?",
  "DQ What is your gender?",
  "DQ What is your academic program?",
  "G39 How often do you eat breakfast during a typical study week?",
  "G39 How many main meals do you usually eat per day, excluding snacks?",
  "G39 How productive do you usually feel during your study sessions?"
)]

# Rename the columns to shorter names
names(group39_data) <- c(
  "age",
  "gender",
  "academic_program",
  "breakfast_frequency",
  "main_meals_per_day",
  "study_productivity"
)

# Convert breakfast frequency to an ordered factor
group39_data <- group39_data %>%
  mutate(
    main_meals_per_day = as.numeric(main_meals_per_day),
    study_productivity = as.numeric(study_productivity),
```

```r
    breakfast_frequency = factor(
      breakfast_frequency,
      levels = c(
        "Never",
        "1-2 days per week",
        "3-4 days per week",
        "5-6 days per week",
        "Every day"
      ),
      ordered = TRUE
    ),
    breakfast_score = as.numeric(breakfast_frequency) - 1
  )

# Show the first rows
head(group39_data)
```

```
##       age gender     academic_program breakfast_frequency main_meals_per_day
## 1   25-30   Male Business Informatics           Every day                  3
## 2   21-23   Male         Data Science   3-4 days per week                  2
## 3   21-23 Female         Data Science           Every day                  3
## 4   25-30 Female         Data Science           Every day                  3
## 5 over 35 Female         Data Science               Never                  2
## 6   25-30   Male         Data Science   1-2 days per week                  1
##   study_productivity breakfast_score
## 1                  3               4
## 2                  3               2
## 3                  3               4
## 4                  4               4
## 5                  3               0
## 6                  5               1
```

The selected dataset contains demographic variables as well as the three variables needed to test the research hypotheses. The variables were converted into appropriate formats for the analysis. The number of main meals per day and study productivity were transformed into numeric variables. Breakfast frequency was converted into an ordered factor because the response categories follow a natural order from "Never" to "Every day". Additionally, a numeric breakfast score was created to use breakfast frequency in the statistical correlation tests.

```r
str(group39_data)
```

```
## 'data.frame':    209 obs. of  7 variables:
##  $ age                : chr  "25-30" "21-23" "21-23" "25-30" ...
##  $ gender             : chr  "Male" "Male" "Female" "Female" ...
##  $ academic_program   : chr  "Business Informatics" "Data Science" "Data Science" "Data Science" ...
##  $ breakfast_frequency: Ord.factor w/ 5 levels "Never"<"1-2 days per week"<..: 5 3 5 5 1 2 3 2 3 2 .
##  $ main_meals_per_day : num  3 2 3 3 2 1 2 3 3 2 ...
##  $ study_productivity : num  3 3 3 4 3 5 3 3 4 3 ...
##  $ breakfast_score    : num  4 2 4 4 0 1 2 1 2 1 ...
```

```r
colSums(is.na(group39_data))
```

```
##                  age               gender     academic_program breakfast_frequency
##                    0                    0                    0                   0
##   main_meals_per_day  study_productivity     breakfast_score
##                    0                    0                    0
```

The dataset contains 209 observations and 7 variables after data preparation. No missing values were found in any of the selected variables, so no observations had to be removed before the analysis. The following sections analyze each research question separately. For each hypothesis, we first use an exploratory visualization, then compute descriptive summary statistics, and finally apply an appropriate statistical test.

Hypothesis 1: Breakfast Frequency and Study Productivity

Research Question: Is the frequency of eating breakfast associated with students' perceived study productivity?

Hypothesis: Students who eat breakfast more often report higher perceived study productivity.

Exploratory Data Analysis

Choose appropriate methods to explore the categorical and quantitative features in your data set (one visualization per research question). Briefly explain the methods and comment on your findings.

Descriptive Inference

Compute suitable summary statistics and display the results in a table.

Analytic Inference

Select an appropriate test statistic to test your hypothesis. Summarize the results and explain how your findings are connected to your research question.

to be added…

Hypothesis 2: Main Meals per Day and Study Productivity

Research Question: Is the number of main meals per day associated with students' perceived study productivity?

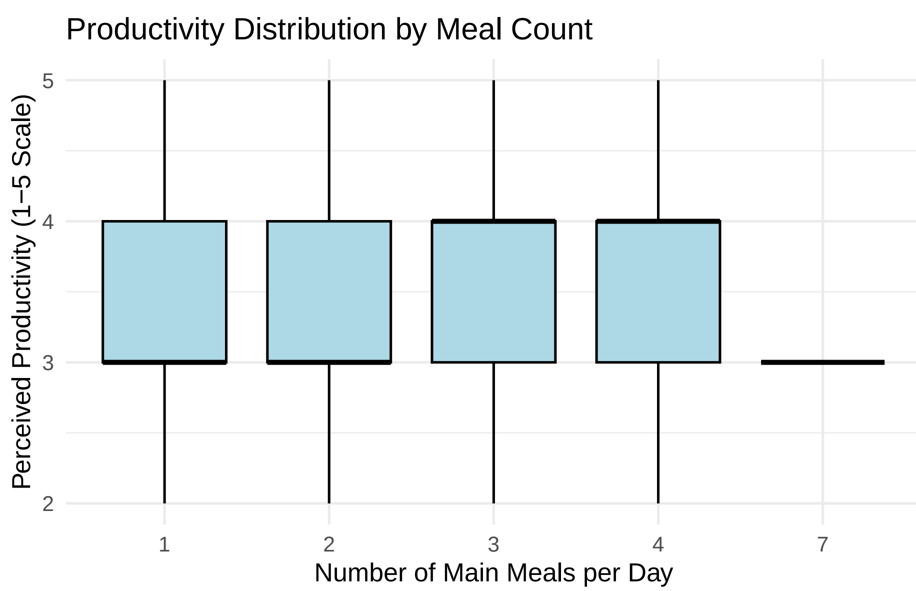
Hypothesis: Students who eat more main meals per day report higher perceived study productivity.

To test this hypothesis, we examine the relationship between two variables: `main_meals_per_day` and `study_productivity`. The variable `main_meals_per_day` is a discrete quantitative variable, as it represents the reported number of main meals students usually eat per day. The variable `study_productivity` is an ordinal variable measured on a 1–5 Likert scale, where higher values indicate higher perceived study productivity.

Exploratory Data Analysis

For the exploratory data analysis, we generate boxplots to visualize the distribution of perceived study productivity scores across the different numbers of reported main meals per day.

```
ggplot(group39_data, aes(x = factor(main_meals_per_day), y = study_productivity)) +
  geom_boxplot(fill = "lightblue", color = "black") +
  labs(
    title = "Productivity Distribution by Meal Count",
    x = "Number of Main Meals per Day",
    y = "Perceived Productivity (1-5 Scale)"
  ) +
  theme_minimal()
```



The boxplot shows that the distributions of perceived study productivity are very similar across students reporting one to four main meals per day. Most productivity scores are between 3 and 4, and the interquartile ranges overlap strongly across these meal-count groups. This suggests that there is no clear visual evidence of a strong association between the number of main meals per day and perceived study productivity.

The category of seven main meals per day was kept in the visualization to reflect the complete dataset. However, this category appears to be an outlier and should be interpreted with caution, as it contains no visible spread in productivity values.

Descriptive Inference

To further examine the pattern observed in the boxplot, we calculate summary statistics for perceived study productivity grouped by the number of main meals per day. This allows us to compare productivity scores across the different meal-count groups.

```r
h2_descriptive <- group39_data %>%
  group_by(main_meals_per_day) %>%
  summarise(
    n = n(),
    mean = round(mean(study_productivity, na.rm = TRUE), 2),
    sd = round(sd(study_productivity, na.rm = TRUE), 2),
    min = min(study_productivity, na.rm = TRUE),
    q25 = round(quantile(study_productivity, 0.25, na.rm = TRUE), 2),
    median = median(study_productivity, na.rm = TRUE),
    q75 = round(quantile(study_productivity, 0.75, na.rm = TRUE), 2),
    max = max(study_productivity, na.rm = TRUE),
    .groups = "drop"
  )

kable(
  h2_descriptive,
  caption = "Summary statistics of perceived study productivity by number of main meals per day",
  col.names = c("Meals", "n", "Mean Productivity", "SD", "Min", "Q25", "Median", "Q75", "Max"),
  booktabs = TRUE
)
```

Table 1: Summary statistics of perceived study productivity by number of main meals per day

Meals	n	Mean Productivity	SD	Min	Q25	Median	Q75	Max
1	17	3.47	1.01	2	3	3	4	5
2	95	3.39	0.72	2	3	3	4	5
3	82	3.52	0.85	2	3	4	4	5
4	14	3.64	0.74	2	3	4	4	5
7	1	3.00	NA	3	3	3	3	3

The descriptive statistics confirm the visual impression from the boxplot. The category of seven main meals per day contains only one observation. Since the category of seven main meals per day contains only one observation and is far outside the range of the other reported meal counts, it was excluded from the inferential test of H2 to avoid giving a single case disproportionate influence. However, it is still shown in the descriptive table to transparently reflect the complete dataset.

Focusing on students reporting one to four main meals per day, perceived study productivity is generally concentrated around values 3 and 4. Students reporting one or two main meals per day have a median productivity score of 3, while students reporting three or four main meals per day have a median productivity score of 4.

The mean productivity scores show a slight increase from 3.47 for students reporting one main meal per day to 3.64 for students reporting four main meals per day. However, the differences between the groups are small, and the quartiles overlap strongly. Therefore, the descriptive statistics suggest at most a weak positive pattern between the number of main meals per day and perceived study productivity.

Analytic Inference

To formally test the hypothesis, we use Spearman's rank correlation coefficient (r_s) as the test statistic. This test is appropriate because perceived study productivity is measured on an ordinal 1–5 Likert scale, while the number of main meals per day is a discrete quantitative variable. Standard parametric correlation tests, such as Pearson's correlation, assume continuous variables, normally distributed data, and a linear relationship. These assumptions are not fully appropriate here because a productivity score of 2 and 3 is not necessarily the same distance apart as a productivity score of 4 and 5. Spearman's correlation is more suitable in this case because it is based on ranks and tests whether there is a monotonic relationship between the two variables.

Since the category of seven main meals per day contains only one observation and was identified as an outlying category in the descriptive analysis, it is excluded from the inferential test.

```r
h2_data <- group39_data %>%
  filter(main_meals_per_day != 7)

spearman_res <- cor.test(
  h2_data$main_meals_per_day,
  h2_data$study_productivity,
  method = "spearman",
  exact = FALSE
)

cat(sprintf("Spearman correlation coefficient: %.3f\n", spearman_res$estimate))
```

```
## Spearman correlation coefficient: 0.091
```

```r
cat(sprintf("p-value: %.4f\n\n", spearman_res$p.value))
```

```
## p-value: 0.1888
```

The Spearman correlation test shows a very small positive correlation between the number of main meals per day and perceived study productivity, $r_s = 0.091$. However, the p-value is $p = 0.1888$, which is higher than the conventional significance level of $\alpha = 0.05$. Therefore, we fail to reject the null hypothesis. This means that the data do not provide sufficient evidence for a statistically significant association between the number of main meals per day and perceived study productivity.

Overall, Hypothesis 2 is not supported by the data.

Hypothesis 3: Breakfast Frequency and Main Meals per Day

Research Question: Is the frequency of eating breakfast associated with the number of main meals students usually eat per day?

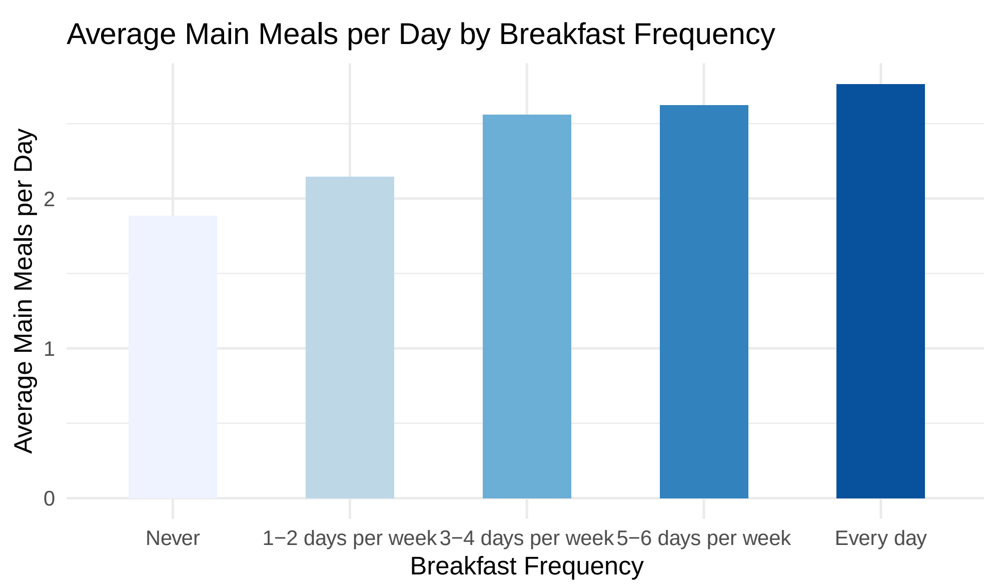
Hypothesis: Students who eat breakfast more often also report eating more main meals per day.

To test this hypothesis, we examine the relationship between two variables: `breakfast_frequency` and `main_meals_per_day`. The variable `breakfast_frequency` is an ordinal variable, as its response categories range from "Never" to "Every day" in a meaningful order. The variable `main_meals_per_day` is a discrete quantitative variable, as it represents the reported number of main meals students usually eat per day.

Exploratory Data Analysis

To explore the relationship between breakfast frequency and the number of main meals per day, we calculated the average number of main meals per day for each breakfast frequency category using the `aggregate()` function. This allows us to observe the central tendency of meal consumption across different breakfast habits. A barplot was created to visualize this relationship, with breakfast frequency on the x-axis and average main meals per day on the y-axis. This visualization helps identify potential trends and patterns in the data.

```r
# Calculate average main meals for each breakfast category
h3_summary <- aggregate(
  main_meals_per_day ~ breakfast_frequency,
  data = group39_data,
  FUN = mean
)

# Create barplot for Hypothesis 3
ggplot(h3_summary, aes(x = breakfast_frequency,
                       y = main_meals_per_day, fill = breakfast_frequency)) +
  geom_bar(stat = "identity", width = 0.5) +
  labs(
    title = "Average Main Meals per Day by Breakfast Frequency",
    x = "Breakfast Frequency",
    y = "Average Main Meals per Day",
    fill = "Breakfast Frequency"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
  ) +
  scale_fill_brewer(palette = "Blues")
```



The exploratory analysis reveals a positive trend between breakfast frequency and the number of main meals consumed per day. Students who never eat breakfast appear to report the lowest average number of main meals per day, while students who eat breakfast every day appear to report the highest average number. The barplot shows a consistent upward pattern: as breakfast frequency increases from "Never" to "Every day", the average number of main meals per day also increases. This initial visual evidence is consistent with the hypothesis that students who eat breakfast more frequently also tend to consume more main meals throughout the day.

Descriptive Inference

To compute the descriptive statistics, we grouped the data by breakfast frequency and calculated the count, mean, standard deviation, minimum, first quartile, median, third quartile, and maximum for the number of main meals per day. This provides an overview of both the central tendency and the dispersion of meal consumption across the different breakfast frequency categories.

```r
# Compute comprehensive summary statistics for main meals per day by breakfast frequency
h3_descriptive <- group39_data %>%
  group_by(breakfast_frequency) %>%
  summarise(
    n = n(),
    mean = round(mean(main_meals_per_day, na.rm = TRUE), 2),
    sd = round(sd(main_meals_per_day, na.rm = TRUE), 2),
    min = min(main_meals_per_day, na.rm = TRUE),
    q25 = round(quantile(main_meals_per_day, 0.25, na.rm = TRUE), 2),
    median = median(main_meals_per_day, na.rm = TRUE),
    q75 = round(quantile(main_meals_per_day, 0.75, na.rm = TRUE), 2),
    max = max(main_meals_per_day, na.rm = TRUE),
    .groups = "drop"
  )

# Display the table
kable(
  h3_descriptive,
  caption = "Summary statistics of main meals per day by breakfast frequency",
  col.names = c("Breakfast Frequency", "n", "Mean", "SD", "Min", "Q25", "Median", "Q75", "Max"),
  booktabs = TRUE
)
```

Table 2: Summary statistics of main meals per day by breakfast frequency

Breakfast Frequency	n	Mean	SD	Min	Q25	Median	Q75	Max
Never	26	1.88	0.65	1	1.25	2	2	3
1-2 days per week	41	2.15	0.48	1	2.00	2	2	3
3-4 days per week	50	2.56	0.95	1	2.00	2	3	7
5-6 days per week	24	2.62	0.92	1	2.00	3	3	4
Every day	68	2.76	0.67	1	2.00	3	3	4

The descriptive statistics confirm the visual impression from the barplot. Students who never eat breakfast report the lowest average number of main meals per day, while students who eat breakfast every day report the highest average number. The mean number of main meals increases consistently across the breakfast frequency categories, from 1.88 among students who never eat breakfast to 2.76 among students who eat breakfast every day.

The median number of main meals is 2 in the lower breakfast-frequency categories and increases to 3 among students who eat breakfast five to six days per week or every day. Although there is some variability within the groups, especially in the "3–4 days per week" and "5–6 days per week" categories, the overall descriptive pattern suggests a positive association between breakfast frequency and the number of main meals per day. Therefore, the descriptive statistics are consistent with H3.

Analytic Inference

To test whether there is a statistically significant association between breakfast frequency and the number of main meals per day, we used Spearman's rank correlation coefficient. This non-parametric test is appropriate because breakfast frequency is an ordinal variable with ordered response categories, and the aim is to assess whether there is a monotonic relationship between breakfast frequency and the number of main meals per day without assuming normally distributed variables.

```r
# Spearman correlation test
h3_correlation <- cor.test(
  group39_data$breakfast_score,
  group39_data$main_meals_per_day,
  method = "spearman",
  use = "complete.obs"
)

# Display results
cat(
  "   SPEARMAN CORRELATION TEST RESULTS - H3\n",
  "Testing: Breakfast Frequency vs Main Meals per Day\n\n",
  sprintf("Correlation coefficient (rho):  %.3f\n", h3_correlation$estimate),
  sprintf("p-value:                        %.4f\n", h3_correlation$p.value),
  sprintf(
    "Sample size (n):                %d\n",
    sum(complete.cases(group39_data$breakfast_score, group39_data$main_meals_per_day))
  ),
  sep = ""
)
```

```
##    SPEARMAN CORRELATION TEST RESULTS - H3
## Testing: Breakfast Frequency vs Main Meals per Day
##
## Correlation coefficient (rho):  0.425
## p-value:                        0.0000
## Sample size (n):                209
```

The Spearman correlation test shows a moderate positive correlation between breakfast frequency and the number of main meals per day, $\rho = 0.425$. The p-value is smaller than 0.001, which is below the conventional significance level of $\alpha = 0.05$. Therefore, we reject the null hypothesis of no association. This means that the data provide evidence for a statistically significant positive association between breakfast frequency and the number of main meals per day. As breakfast frequency increases, the number of main meals per day also tends to increase.

Overall, Hypothesis 3 is supported by the data. Students who eat breakfast more frequently tend to report eating more main meals per day.

Explorative Analysis

Mediator Hypotheses: Demographic Questions as Moderators and Mediators

We explore whether the demographic variables (age, gender, academic program) moderate or mediate the relationships between meal habits and study productivity. Specifically, we test whether the associations between breakfast frequency and productivity, and between main meals per day and productivity, differ across demographic groups.

Moderation by Demographics

Gender as a Moderator We test whether the relationship between main meals per day and study productivity differs by gender using an interaction model.

```
# Prepare gender as a factor (excluding "Other" and "Prefer not to say" if present)
group39_data_gender <- group39_data %>%
  filter(gender %in% c("Female", "Male")) %>%
  mutate(gender = factor(gender))

# Interaction model: main meals per day × gender -> study productivity
interaction_model_gender <- lm(
  study_productivity ~ main_meals_per_day * gender,
  data = group39_data_gender
)

summary(interaction_model_gender)
```

```
##
## Call:
## lm(formula = study_productivity ~ main_meals_per_day * gender,
##     data = group39_data_gender)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.5217 -0.4451 -0.3599  0.5592  1.6401
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 3.59651    0.41687   8.627 1.73e-15 ***
## main_meals_per_day         -0.07571    0.16783  -0.451    0.652
## genderMale                 -0.31750    0.46224  -0.687    0.493
## main_meals_per_day:genderMale  0.15662    0.18418   0.850    0.396
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7992 on 205 degrees of freedom
## Multiple R-squared:  0.007784,   Adjusted R-squared:  -0.006737
## F-statistic: 0.536 on 3 and 205 DF,  p-value: 0.6581
```

```
# Visualize the interaction: separate regression lines by gender
ggplot(group39_data_gender, aes(x = main_meals_per_day, y = study_productivity, color = gender)) +
  geom_point(alpha = 0.4) +
```

```r
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "Main meals per day",
    y = "Study productivity (1-5)",
    title = "Main meals per day -> Study productivity by gender",
    color = "Gender"
  ) +
  theme_minimal()
```

If the interaction term is significant, the relationship between meal frequency and productivity differs by gender, indicating gender as a moderator.

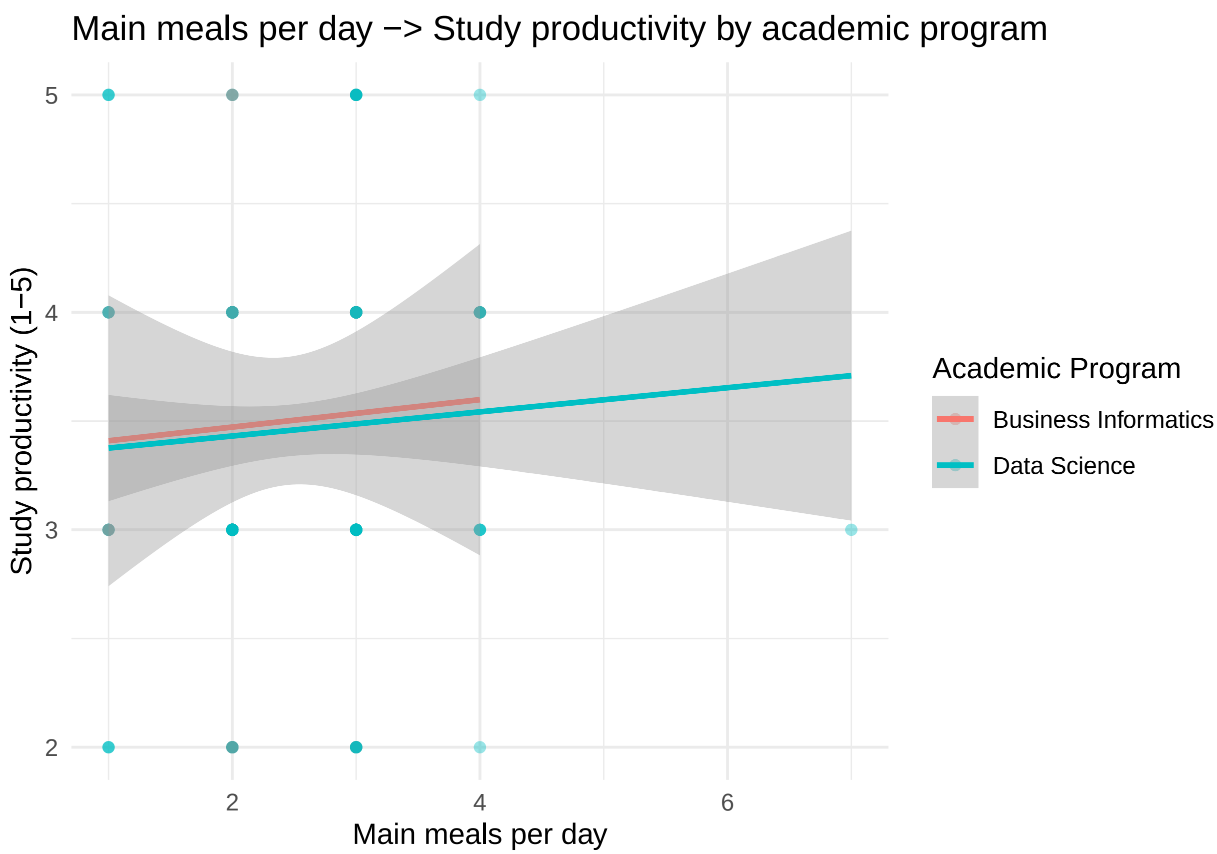
```r
# Interaction model: main meals per day × academic program -> study productivity
interaction_model_program <- lm(
  study_productivity ~ main_meals_per_day * academic_program,
  data = group39_data
)

summary(interaction_model_program)
```

Academic Program as a Moderator

```
##
```

```
## Call:
## lm(formula = study_productivity ~ main_meals_per_day * academic_program,
##     data = group39_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.5423 -0.4868 -0.3757  0.5276  1.6242
##
## Coefficients:
##                                                   Estimate Std. Error t value
## (Intercept)                                       3.346457   0.510030   6.561
## main_meals_per_day                                0.062992   0.201019   0.313
## academic_programData Science                     -0.026227   0.544983  -0.048
## main_meals_per_day:academic_programData Science -0.007468   0.214100  -0.035
##                                                  Pr(>|t|)
## (Intercept)                                      4.27e-10 ***
## main_meals_per_day                                  0.754
## academic_programData Science                        0.962
## main_meals_per_day:academic_programData Science    0.972
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8009 on 205 degrees of freedom
## Multiple R-squared:  0.003605,   Adjusted R-squared:  -0.01098
## F-statistic: 0.2473 on 3 and 205 DF,  p-value: 0.8632
```

```
# Visualize the interaction: separate regression lines by program
ggplot(group39_data, aes(x = main_meals_per_day, y = study_productivity, color = academic_program)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "Main meals per day",
    y = "Study productivity (1-5)",
    title = "Main meals per day -> Study productivity by academic program",
    color = "Academic Program"
  ) +
  theme_minimal()
```



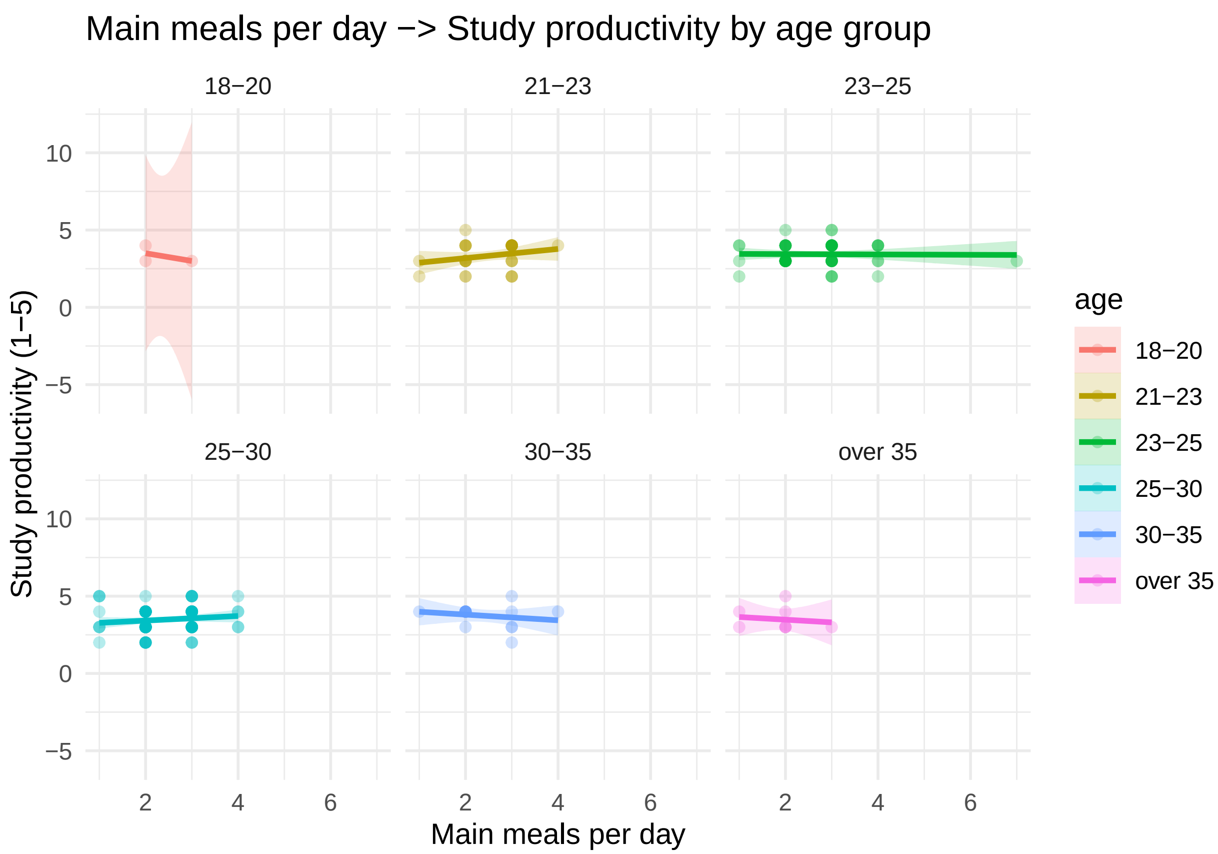

```r
# Interaction model: main meals per day × age -> study productivity
interaction_model_age <- lm(
  study_productivity ~ main_meals_per_day * age,
  data = group39_data
)

summary(interaction_model_age)
```

Age as a Moderator

```
##
## Call:
## lm(formula = study_productivity ~ main_meals_per_day * age, data = group39_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.6228 -0.4438  0.0000  0.5670  1.8148
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    4.5000     2.3452   1.919   0.0565 .
## main_meals_per_day            -0.5000     0.9852  -0.508   0.6124
## age21-23                      -1.9074     2.4127  -0.791   0.4302
```

```
## age23-25                       -1.0345     2.3662  -0.437   0.6624
## age25-30                       -1.3740     2.3626  -0.582   0.5615
## age30-35                       -0.3246     2.4564  -0.132   0.8950
## ageover 35                     -0.6739     2.5243  -0.267   0.7898
## main_meals_per_day:age21-23     0.7963     1.0092   0.789   0.4311
## main_meals_per_day:age23-25     0.4892     0.9913   0.493   0.6222
## main_meals_per_day:age25-30     0.6475     0.9918   0.653   0.5146
## main_meals_per_day:age30-35     0.3158     1.0275   0.307   0.7589
## main_meals_per_day:ageover 35   0.3261     1.0935   0.298   0.7659
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8044 on 197 degrees of freedom
## Multiple R-squared:  0.03415,    Adjusted R-squared:  -0.01979
## F-statistic: 0.6331 on 11 and 197 DF,  p-value: 0.7991
```

```r
# Visualize the moderation: productivity by meals, colored by age group
ggplot(group39_data, aes(x = main_meals_per_day, y = study_productivity, color = age)) +
  geom_point(alpha = 0.3) +
  facet_wrap(~ age) +
  geom_smooth(method = "lm", se = TRUE, aes(fill = age), alpha = 0.2) +
  labs(
    x = "Main meals per day",
    y = "Study productivity (1-5)",
    title = "Main meals per day -> Study productivity by age group"
  ) +
  theme_minimal()
```



Mediation via Breakfast Frequency

We test whether the relationship between breakfast frequency and study productivity is mediated by the number of main meals per day. The hypothesized pathway is:

breakfast_frequency -> main_meals_per_day -> study_productivity

```
# Step 1: Total effect (breakfast -> productivity)
model_total <- lm(
  study_productivity ~ breakfast_score,
  data = group39_data
)

# Step 2: Effect of breakfast on mediator (breakfast -> meals)
model_mediator <- lm(
  main_meals_per_day ~ breakfast_score,
  data = group39_data
)

# Step 3: Direct and indirect effects (breakfast + meals -> productivity)
model_full <- lm(
  study_productivity ~ breakfast_score + main_meals_per_day,
  data = group39_data
)

cat("Total effect (c):\n")
```

```
## Total effect (c):
```

```r
print(coef(model_total)["breakfast_score"])
```

```
## breakfast_score
##      0.09988595
```

```r
cat("\nEffect on mediator (a):\n")
```

```
##
## Effect on mediator (a):
```

```r
print(coef(model_mediator)["breakfast_score"])
```

```
## breakfast_score
##       0.2087705
```

```r
cat("\nDirect effect (c'):\n")
```

```
##
## Direct effect (c’):
```

```r
print(coef(model_full)["breakfast_score"])
```

```
## breakfast_score
##       0.1021071
```

```r
cat("\nMediator effect (b):\n")
```

```
##
## Mediator effect (b):
```

```r
print(coef(model_full)["main_meals_per_day"])
```

```
## main_meals_per_day
##        -0.01063931
```

```r
cat("\nIndirect effect (a × b):\n")
```

```
##
## Indirect effect (a × b):
```

```r
a <- coef(model_mediator)["breakfast_score"]
b <- coef(model_full)["main_meals_per_day"]
indirect <- a * b
print(indirect)
```

```
## breakfast_score
##    -0.002221174
```

```r
cat("\nProportion of total effect mediated:\n")
```

```
##
## Proportion of total effect mediated:
```

```r
total_c <- coef(model_total)["breakfast_score"]
prop_mediated <- indirect / total_c
print(prop_mediated)
```

```
## breakfast_score
##      -0.0222371
```

```r
# Simple conceptual diagram
dag_data <- tibble(
  label = c("Breakfast\nFrequency", "Main Meals\nper Day", "Study\nProductivity"),
  x = c(0, 1, 2),
  y = c(0, 0.5, 0)
)

# Alternative: use a summary table of the pathways
mediation_summary <- tibble(
  Pathway = c(
    "Total effect (c): breakfast -> productivity",
    "Path a: breakfast -> main meals",
    "Path b: main meals -> productivity (controlling for breakfast)",
    "Direct effect (c'): breakfast -> productivity (controlling for meals)",
    "Indirect effect (a × b): mediated through meals"
  ),
  Coefficient = c(
    coef(model_total)["breakfast_score"],
    coef(model_mediator)["breakfast_score"],
    coef(model_full)["main_meals_per_day"],
    coef(model_full)["breakfast_score"],
    a * b
  )
)

kable(
  mediation_summary,
  caption = "Mediation analysis: breakfast frequency -> main meals -> study productivity"
)
```

Table 3: Mediation analysis: breakfast frequency -> main meals -> study productivity

Pathway	Coefficient
Total effect (c): breakfast -> productivity	0.0998859
Path a: breakfast -> main meals	0.2087705
Path b: main meals -> productivity (controlling for breakfast)	-0.0106393
Direct effect (c'): breakfast -> productivity (controlling for meals)	0.1021071

Pathway	Coefficient
Indirect effect (a × b): mediated through meals	-0.0022212

If the indirect effect (a × b) is significant and substantially reduces the direct effect (c -> c'), this indicates that main meals per day mediates the relationship between breakfast frequency and study productivity.

Subgroup Analysis: Descriptive Statistics by Demographics

```r
# Subgroup analysis: mean productivity by gender and meal consumption
subgroup_gender <- group39_data %>%
  filter(gender %in% c("Female", "Male")) %>%
  group_by(gender) %>%
  summarise(
    n = n(),
    mean_meals = round(mean(main_meals_per_day), 2),
    mean_productivity = round(mean(study_productivity), 2),
    .groups = "drop"
  )

kable(subgroup_gender, caption = "Mean main meals and study productivity by gender")
```

Table 4: Mean main meals and study productivity by gender

gender	n	mean_meals	mean_productivity
Female	53	2.40	3.42
Male	156	2.49	3.48

```r
# Subgroup analysis: mean productivity by academic program
subgroup_program <- group39_data %>%
  group_by(academic_program) %>%
  summarise(
    n = n(),
    mean_meals = round(mean(main_meals_per_day), 2),
    mean_productivity = round(mean(study_productivity), 2),
    .groups = "drop"
  )

kable(subgroup_program, caption = "Mean main meals and study productivity by academic program")
```

Table 5: Mean main meals and study productivity by academic program

academic_program	n	mean_meals	mean_productivity
Business Informatics	32	2.44	3.50
Data Science	177	2.47	3.46

```r
# Subgroup analysis: mean productivity by age group
subgroup_age <- group39_data %>%
  group_by(age) %>%
  summarise(
    n = n(),
    mean_meals = round(mean(main_meals_per_day), 2),
    mean_productivity = round(mean(study_productivity), 2),
    .groups = "drop"
  )

kable(subgroup_age, caption = "Mean main meals and study productivity by age group")
```

Table 6: Mean main meals and study productivity by age group

age	n	mean_meals	mean_productivity
18-20	3	2.33	3.33
21-23	30	2.50	3.33
23-25	55	2.69	3.44
25-30	98	2.40	3.48
30-35	15	2.40	3.73
over 35	8	1.88	3.50

Interpretation

The interaction models reveal whether demographic characteristics moderate the meal-productivity relationship. Significant interactions indicate that the strength or direction of the association differs across demographic groups. The mediation analysis explores whether the effect of breakfast frequency on productivity operates through an increase in main meals consumed. Subgroup descriptive statistics provide context for interpreting potential moderation effects.

Summary of Findings

Conclusion